

Targeted Drill — Day 1 Weak Areas

Five concepts you missed, re-drilled

Wednesday, June 17, 2026 • 15 questions • derivatives · decay & half-life · trapezoidal rule · limits & small-angle · Newton–Raphson

How to use it: Work each block with the FE Reference Handbook open — these are the exact ideas that cost you points on Day 1. Aim for ≈ 3 min/question, then study every worked solution and the handbook section it cites. All of this lives in the handbook's *Mathematics* section.

Questions

Derivatives (power rule)

- For $f(x) = 2x^4 - 3x^2 + 5x - 7$, the derivative $f'(1) =$
(A) 4 (B) 7
(C) 10 (D) 13
- The slope of the line tangent to $y = x^3 - 4x$ at $x = 2$ is:
(A) 4 (B) 6
(C) 8 (D) 12
- The derivative $\frac{d}{dx}[\ln x]$ evaluated at $x = 4$ is:
(A) 0.25 (B) 0.5
(C) 1 (D) 4

First-order decay & half-life

- A contaminant decays first order with $k = 0.20 \text{ day}^{-1}$. Its half-life is closest to:
(A) 1.4 days (B) 3.5 days
(C) 5.0 days (D) 6.9 days
- A radionuclide has a half-life of 14 days. Its first-order decay constant k is closest to:
(A) 0.025 day^{-1} (B) 0.050 day^{-1}
(C) 0.099 day^{-1} (D) 0.139 day^{-1}
- A first-order reaction has $k = 0.05 \text{ hr}^{-1}$. The time to remove 75% of the compound is closest to:
(A) 13.9 hr (B) 20.8 hr
(C) 27.7 hr (D) 55.4 hr

Trapezoidal rule (numerical integration)

- Using the trapezoidal rule with $h = 2$, estimate $\int_0^4 f dx$ given $f(0) = 0$, $f(2) = 4$, $f(4) = 16$.
(A) 21.3 (B) 24
(C) 32 (D) 40

8. A vehicle's speed is recorded as $v(0) = 0$, $v(3) = 9$, $v(6) = 12$ (m/s at t in s). Using the trapezoidal rule, the distance traveled over 6 s is closest to:
- (A) 36 m (B) 42 m
(C) 45 m (D) 63 m
9. Using the trapezoidal rule with $h = 1$, estimate $\int_1^3 \frac{1}{x} dx$ (use $f = 1, 0.5, 0.333$).
- (A) 0.92 (B) 1.10
(C) 1.17 (D) 1.33

Limits & small-angle approximation

10. $\lim_{x \rightarrow 0} \frac{\sin 3x}{x} =$
- (A) 0 (B) 1
(C) 3 (D) ∞
11. $\lim_{x \rightarrow 0} \frac{\sin 5x}{2x} =$
- (A) 1 (B) 2.5
(C) 5 (D) 10
12. Using the small-angle approximation, $\tan(0.04 \text{ rad}) \approx$
- (A) 0.0004 (B) 0.04
(C) 0.40 (D) 2.29

Newton–Raphson method

13. At each step, the Newton–Raphson method requires:
- (A) the derivative $f'(x)$ (B) two points bracketing the root
(C) a matrix inverse (D) numerical integration
14. For $f(x) = x^2 - 2$ with starting guess $x_0 = 1$, one Newton–Raphson iteration gives $x_1 =$
- (A) 1.25 (B) 1.33
(C) 1.50 (D) 2.00
15. For $f(x) = x^2 - 10$ with $x_0 = 3$, one Newton–Raphson iteration gives $x_1 \approx$
- (A) 3.000 (B) 3.083
(C) 3.167 (D) 3.333

Answer Key

1	2	3	4	5	6	7	8
B	C	A	B	B	C	B	C
9	10	11	12	13	14	15	
C	C	B	B	A	C	C	

Worked Solutions

1. **B.** $f'(x) = 8x^3 - 6x + 5$; at $x = 1$: $8 - 6 + 5 = 7$. (Don't drop the $+5x$ term — its derivative is 5.) Handbook: Differential Calculus — power rule.
2. **C.** The tangent slope is the derivative: $\frac{dy}{dx} = 3x^2 - 4$; at $x = 2$: $12 - 4 = 8$. Handbook: Differential Calculus — slope/tangent.
3. **A.** $\frac{d}{dx} \ln x = \frac{1}{x}$; at $x = 4$: $\frac{1}{4} = 0.25$. Handbook: Differential Calculus — derivatives of transcendental functions.
4. **B.** $t_{1/2} = \frac{0.693}{k} = \frac{0.693}{0.20} = 3.47 \approx 3.5$ days. Handbook: Differential Equations — first-order decay.
5. **B.** Rearrange the same formula: $k = \frac{0.693}{t_{1/2}} = \frac{0.693}{14} = 0.0495 \approx 0.050 \text{ day}^{-1}$. Handbook: Differential Equations — first-order decay.
6. **C.** Removing 75% leaves 25% — exactly two half-lives. $t = 2 t_{1/2} = 2\left(\frac{0.693}{0.05}\right) = 2(13.86) = 27.7$ hr. (General form: $t = \frac{\ln(C_0/C)}{k} = \frac{\ln 4}{0.05} = 27.7$ hr.) Handbook: Differential Equations / reaction kinetics.
7. **B.** Trapezoidal rule: $\frac{h}{2}[f_0 + 2f_1 + f_2] = \frac{2}{2}[0 + 2(4) + 16] = 24$. (Exact $\int_0^4 x^2 dx = 21.3$; the rule runs slightly high.) Handbook: Numerical Methods — numerical integration.
8. **C.** Distance $= \int v dt \approx \frac{h}{2}[v_0 + 2v_1 + v_2] = \frac{3}{2}[0 + 2(9) + 12] = \frac{3}{2}(30) = 45$ m. Handbook: Numerical Methods — numerical integration.
9. **C.** $\frac{h}{2}[f_0 + 2f_1 + f_2] = \frac{1}{2}[1 + 2(0.5) + 0.333] = \frac{1}{2}(2.333) = 1.17$. (Exact $= \ln 3 = 1.10$; again a slight overestimate.) Handbook: Numerical Methods — numerical integration.
10. **C.** For small x , $\sin 3x \approx 3x$, so $\frac{\sin 3x}{x} \rightarrow 3$. (Or L'Hôpital: $\frac{3 \cos 3x}{1} \rightarrow 3$.) Handbook: Differential Calculus — limits / L'Hôpital.
11. **B.** $\sin 5x \approx 5x$, so $\frac{\sin 5x}{2x} \rightarrow \frac{5x}{2x} = \frac{5}{2} = 2.5$. Handbook: Differential Calculus — limits / L'Hôpital.
12. **B.** Small-angle approximation (radians): $\tan \theta \approx \theta$, so $\tan(0.04) \approx 0.04$. (Option D, 2.29, is 0.04 rad expressed in degrees — a units trap.) Handbook: Trigonometry — small-angle approximation.
13. **A.** Each iteration $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ needs the derivative $f'(x)$. (Bracketing two points is the *bisection* method, not Newton-Raphson.) Handbook: Numerical Methods — root finding.
14. **C.** $f'(x) = 2x$, so $x_1 = 1 - \frac{1^2 - 2}{2(1)} = 1 - \frac{-1}{2} = 1.50$. (Heading toward $\sqrt{2} = 1.414$.) Handbook: Numerical Methods — root finding.

15. **C.** $x_1 = 3 - \frac{3^2 - 10}{2(3)} = 3 - \frac{-1}{6} = 3.167$. (Heading toward $\sqrt{10} = 3.162$.)
Numerical Methods — root finding.

Handbook:

Pattern to remember: half-life is always $0.693/k$; trapezoidal weights interior points twice; $\sin(ax)/x \rightarrow a$; Newton–Raphson slides down the tangent and needs $f'(x)$.